

U2210251

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$$f(x) = 2x^3 + 3x^2 - 36x$$

$$f'(x) = 6x^2 + 6x - 36$$

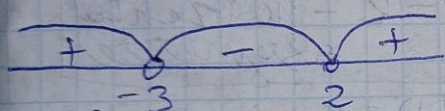
(a):

$$f'(x) = 6x^2 + 6x - 36 = 0$$

$$x^2 + x - 6 = 0$$

$$D = \sqrt{1 + 4 \cdot 6} = 5$$

$$x_{1,2} = \frac{-1 \pm 5}{2} = \begin{cases} -3 \\ 2 \end{cases}$$



$$(-4) \cdot 6 + 6 \cdot (-4) - 36 =$$

$$= 16 \cdot 6 - 24 - 36 =$$

$$= 96 - 60 = 36 > 0$$

$$6 \cdot 0 + 6 \cdot 0 - 36 =$$

$$= -36 < 0$$

$$6 \cdot (3)^2 + 6 \cdot (3) - 36 =$$

$$= 6(9 + 3) - 3(12) =$$

$$= 12(6 - 3) = 36 > 0$$

$f(x)$ is up at $(-\infty; -3)$
and $(2; \infty)$, and down
at $(-3; 2)$.

(b):

$$f'(x) = 0$$

$$x^2 + x - 6 = 0$$

$$x_1 = -3$$

$$x_2 = 2$$

$$f(x) = 2x^3 + 3x^2 - 36x$$

$$f(-3) = -54 + 27 + 36 \cdot 3 =$$

$$= 108 - 27 = 81$$

$$f(2) = 16 + 12 - 72 =$$

$$= -44$$

~~maximum:~~

~~$f(x)$ is rising until
 $x = -3$, and then went
down, maximum is
equal to: $x = -3, f(x) = 81$~~

as $f(x) > 0$ until $x = -3$
then $f(x) < 0$, $x = -3$
~~is local maximum~~
and $f(x) = 81$ is local
maximum.

as $f(x) < 0$ until $x = 2$
then $f(x) > 0$, $x = 2$, $f(x) = -44$
is local minimum.

(C):

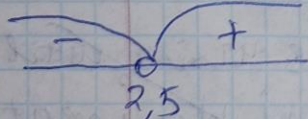
$$f(x) = 2x^3 + 3x^2 - 36x$$

$$f'(x) = 6x^2 + 6x - 36$$

$$f''(x) = 12x + 6 - 36$$

$$12x + 6 - 36 = 0$$

$$x = 2,5$$



on the interval
 $x \in (-\infty; 2,5)$ it is
concave downward
and on the interval

$x \in (2,5; \infty)$ it is
concave upward

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$$y = \frac{1+2x^2}{1+x^2}$$

$$y + yx^2 = 1 + 2x^2$$

$$y - 1 = x^2(2 - y)$$

$$y = 2 - \text{horizontal asymptote}$$

$$f(x) = y$$

$$\cancel{f'(x) = 4x \cdot (1+x^2)}$$

$$f'(x) = \frac{4x(1+x^2) - 2x(1+2x^2)}{1+2x^2+x^4}$$

$$= \frac{4x - 2x + 4x^3 - 4x^3}{1+2x^2+x^4} =$$

$$= \frac{2x}{1+2x^2+x^4}$$

$$\frac{2x}{1+2x^2+x^4} = 0$$

$$x_1 = 0$$

$$1+2x^2+x^4 \neq 0$$

$$x^2 = \pm$$

$$1+2\pm+\pm^2 \neq 0$$

$$D = 4 - 4 = 0$$

$$\pm = -\frac{2}{2} = -1$$

$$x \in \emptyset$$



$$f'(x) > 0$$

when $x > 0$

$$\text{and } f'(x) < 0$$

when $x < 0$

$$f'(0) = 0$$

$$f(0) = 1$$

$f(x)$ has local

minimum at $x=0$; $f(x)=1$

$$f''(x) = 2(1+2x^2+x^4) - 2x(4x+4x^3)$$

$$f''(x) = \frac{2-4x^2-6x^4}{(1+x^2)^4}$$

$$(1+x^2)^4 > 0 \text{ and } (1+x^2)^4 \neq 0$$

$$(1+x^2)^4 > 0 \text{ and } (1+x^2)^4 \neq 0$$

$$2 - 4x^2 - 6x^4 = 0$$

$$3x^4 + 2x^2 - 1 = 0$$

$$x^2 = t$$

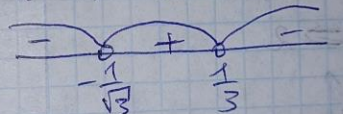
$$3t^2 + 2t - 1 = 0$$

$$t = \frac{-2 \pm \sqrt{4 + 4 \cdot 3}}{6} = \begin{cases} t_1 = -1 \\ t_2 = \frac{1}{3} \end{cases}$$

$$x^2 = -1 \Rightarrow x \in \emptyset$$

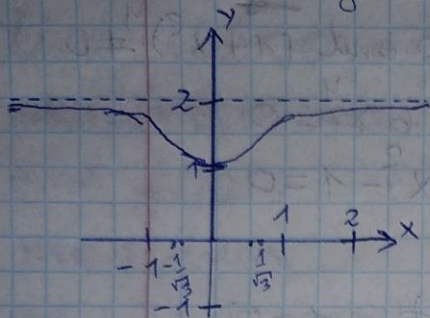
$$x^2 = \frac{1}{3} \Rightarrow x = \pm \frac{1}{\sqrt{3}}$$

$$2 - 4x^2 - 6x^4 = 0$$



$f''(0) > 0$ which proves that $f(x)$ has minpoint at 0

So the rough sketch is



if we put some values

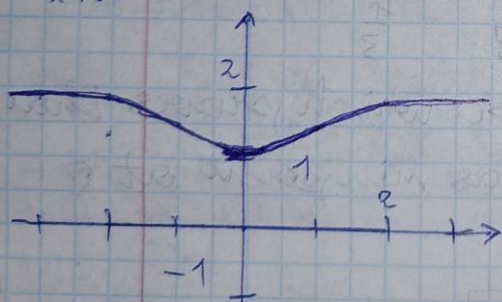
$$f(1) = 1,5 = f(-1)$$

$$f(0,5) = f(-0,5) = 1,2$$

$$f(-2) = f(2) = 1,8$$

$$f(-3) = f(3) = 1,9$$

$$\lim_{x \rightarrow \infty} f(x) = 2$$



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$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1; y = \pm \frac{b}{a}x$$

$$\frac{x^2}{a^2} - 1 = \frac{y^2}{b^2}$$

$$b^2 \left(\frac{x^2}{a^2} - \frac{a^2}{a^2} \right) = y^2$$

$$y^2 = \frac{b^2}{a^2} (x^2 - a^2)$$

$$y = \pm \frac{b}{a} \sqrt{x^2 - a^2}$$

$$\lim_{x \rightarrow \infty} \left[\frac{b}{a} \sqrt{x^2 - a^2} - \frac{b}{a} x \right] = \frac{b}{a} \lim_{x \rightarrow \infty} \left[\sqrt{x^2 - a^2} - x \right] =$$
$$= \frac{b}{a} \lim_{x \rightarrow \infty} \left[\left(\sqrt{x^2 - a^2} - x \right) \cdot \frac{\sqrt{x^2 - a^2} + x}{\sqrt{x^2 - a^2} + x} \right] =$$

$$= \frac{b}{a} \lim_{x \rightarrow \infty} \frac{-a^2}{\sqrt{x^2 - a^2} + x} = 0$$

if $y = -\frac{b}{a}x$

$$\lim_{x \rightarrow \infty} \left[-\frac{b}{a} \sqrt{x^2 - a^2} + \frac{b}{a} x \right] = -\frac{b}{a} \lim_{x \rightarrow \infty} \frac{-a^2}{\sqrt{x^2 - a^2} + x} = 0$$

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Page 267 / Ex 13

$$f(x) = x^4 - 2x^3 + 5x^2 - 6 = 0, x \in [1, 2]$$

$$f'(x) = 4x^3 - 6x^2 + 10x$$

$$x_{n+1} = x_n - \frac{x_n^4 - 2x_n^3 + 5x_n^2 - 6}{4x_n^3 - 6x_n^2 + 10x_n}$$

$$x_2 = 2 - \frac{2^4 - 2 \cdot 2^3 + 5 \cdot 2^2 - 6}{4 \cdot 2^3 - 6 \cdot 2^2 + 10 \cdot 2} = \frac{3}{2}$$

$$x_3 = \frac{3}{2} - \frac{1,5^4 - 2 \cdot 1,5^3 + 5 \cdot 1,5^2 - 6}{4 \cdot 1,5^3 - 6 \cdot 1,5^2 + 10 \cdot 1,5} = 1,2625$$

$$x_4 = 1,2625 - \frac{1,2625^4 - 2 \cdot 1,2625^3 + 5 \cdot 1,2625^2 - 6}{4 \cdot 1,2625^3 - 6 \cdot 1,2625^2 + 10 \cdot 1,2625} =$$
$$= 1,218807; x_5(1,218807) = 1,214563$$

$$x_6(1,214563) = 1,214562$$

$$x_7(1,214562) = 1,214562$$